

Startup Run 3D: Entrepreneurship Simulator

TARGET ENGINE
Three.js / WebGL

FILE STRATEGY
Single-File HTML

VISUAL STYLE
3D Toon / Comic

STATE STORAGE
Local Storage API

1. AI Model Handoff Architecture

PROTOCOL

This master specification is optimized for seamless transfer across AI platforms (Claude, ChatGPT, Gemini). It ensures complete code continuity without feature degradation when switching models or hitting token boundaries.

Single-File Execution Rule

During development, all HTML, CSS styling, Three.js 3D rendering, game loop physics, financial balance formulas, and leaderboard code MUST remain strictly contained within a single index.html file. File splitting occurs only after full game stability is achieved.

2. Universal Model Handoff Prompt Template

COPY & PASTE

Copy the following block directly into a new AI model whenever you transition development tasks:

```
System Context: You are lead game developer building "Startup Run 3D" in a SINGLE HTML file (HTML5 + Three.js + Inline CSS). Current Specification Baseline: 1. Engine: Three.js (via CDN), WebGL low-poly toon rendering (MeshToonMaterial). 2. Cutscene: NYC Apartment room, camera outside looking in through window, subtitle overlay with startup intro text. 3. Core Loop: Endless runner. Balance ($) constantly decreases (burn rate). Left/Right keys steer through green (buff) and red (debuff) financial gates. 4. Metrics: Survival Timer (MM:SS) + Top-10 Local Storage Leaderboard. Current Working Code: [INSERT EXISTING INDEX.HTML HERE OR STATE "Generate Fresh Base"] Task: Implement [INSERT SPECIFIC FEATURE / BUG FIX] without breaking existing functionality. Maintain single-file HTML structure.
```

3. Technical Stack & Dependencies

TECH STACK

MODULE	TECHNOLOGY	IMPLEMENTATION METHOD
3D Rendering	Three.js (r128)	Single CDN Script Tag (`https://cdnjs.cloudflare.com/ajax/libs/three.js/r128/three.min.js`)
UI & Cutscene	CSS3 Overlay & Flexbox	Absolute overlay positioned above the WebGL Canvas with high `z-index`
Comic Typography	Google Fonts (Bangers / Inter)	Embedded web fonts for comic-book style dialogue boxes
Leaderboard	Browser LocalStorage API	JSON array parsing storing Top 10 longest survival times (`MM:SS`)

4. Stage 1: Comic-Book Cutscene Specification

INTRO SCENE

The game opens in a frozen comic-book panel state representing a classic New York City apartment startup office.

Startup Naming Phase

- Simple overlay modal asking: "Enter Your Startup Name:"
- Input field with default placeholder: Unicorn AI Corp
- "Launch Startup" button initializes the cutscene sequence.

Comic Cutscene Presentation

- Thick black border CSS frame around screen (Comic Panel effect).
- Camera positioned outside looking inside through a window grid.
- 3D Character styled sweeping/cleaning a messy apartment room.

Mandatory Intro Subtitle Dialogue

"Ah, beautiful morning for the start of your startup business, hope you can manage to keep your business running as there is tons of competition."

5. Stage 2: Core Gameplay & Burn-Rate Physics

GAMEPLAY

Upon clicking "Start Running", the 3D camera smoothly transitions from the window angle to an over-the-shoulder third-person runner camera view.

Movement & Lane Controls

- Track Layout:** 3 Lanes (Left, Center, Right).
- Controls:** Arrow Keys (Left/Right) or A/D keys.
- Speed Acceleration:** Base forward speed increases by 2% every 10 seconds survived.

Financial Burn-Rate Engine

- Initial Capital:** \$50,000 Starting Account Balance.
- Passive Burn Rate:** -\$1,000 per second (increases with game time).
- Game Over Trigger:** Balance reaches \$0 or negative value.

6. Stage 3: Buff & Debuff Decision Gates

GATE MATRIX

Pairs of 3D gates spawn endlessly along the track. The player must choose which gate to run through based on their current balance and financial strategy.

GATE TYPE	GATE TITLE TEXT	FINANCIAL IMPACT FORMULA	VISUAL STYLE
BUFF	Angel Seed Investment	+ \$25,000 Balance	Green Neon Border + Floating Particles
BUFF	Product Goes Viral	+ \$15,000 Balance & Reduces Burn Rate by 10%	Green Neon Border + Sparkle Effect
DEBUFF	AWS Server Outage	- \$20,000 Balance	Red Glowing Border + Hazard Stripes
DEBUFF	Unexpected Tax Audit	- \$35,000 Balance	Red Glowing Border + Alarm Flicker
TRADE-OFF	Hire Top Executive	- \$10,000 Instant / + \$500/sec long-term income	Yellow Border + Executive Icon

7. Survival Timer & Top-10 High Score System

LEADERBOARD

Player survival duration is measured continuously from the moment forward running begins until the balance reaches \$0.

Survival Timer HUD

- Real-time HUD counter displayed at top center:
SURVIVAL TIME: 02:45
- Timer pauses on game over and formats final time into seconds for ranking logic.

Top-10 Local Storage Ranking

- Stores an array of objects: [{ companyName, survivalTime, finalBalance }]
- Automatically sorts entries by longest survival time descending.
- Maintains strictly the top 10 historical records in browser storage.

8. Single-File Code Generation Reference

CODE STRUCTURE

When prompting the AI to write or edit the codebase, instruct it to strictly maintain the following internal structure inside index.html:

```
<!DOCTYPE html> <html lang="en"> <head> <style> /* 1. Reset & Fullscreen Canvas Styles
*/ /* 2. Comic Panel Overlay & Cutscene Speech Bubbles */ /* 3. HUD (Balance, Timer) &
Leaderboard Modal Styling */ </style> <script
src="https://cdnjs.cloudflare.com/ajax/libs/three.js/r128/three.min.js"></script> </head>
<body> <!-- HTML UI: Naming Modal, Comic Speech Subtitle Box, HUD, Game Over Screen -->
<script> // MODULE 1: Game State Variables (balance, time, companyName, highScores) //
MODULE 2: Three.js Setup (Scene, Window Camera, Player Mesh, Toon Lighting) // MODULE 3:
Intro Cutscene Logic & Camera Transition // MODULE 4: Endless Track, Obstacle & Gate
Spawning Loop // MODULE 5: Collision Detection & Balance Updates // MODULE 6: Timer Logic &
LocalStorage Top-10 Ranking Engine </script> </body> </html>
```

9. Future Refactoring Blueprint (Multi-File Split)

MODULARIZING

Once the game is 100% polished and fully bug-free in single-file format, instruct the AI to extract the code into the following modular file directory:

TARGET FILE	EXTRACTED RESPONSIBILITIES
index.html	HTML structure, UI modals, HUD overlays, CDN imports
css/style.css	Comic overlays, font definitions, HUD positioning, responsive layout
js/scene.js	Three.js scene initialization, lighting, toon materials, camera controller
js/runner.js	Player movement physics, track generation, gate collision detection
js/gameplay.js	Financial burn-rate math, buff/debuff formulas, game loop management
js/leaderboard.js	LocalStorage reading/writing, Top-10 sorting, survival timer management